

LABORATORY TECHNICAL REPORT

EXPERIMENT 06

Time-Series Analysis and Forecasting of Reported Crime Incidents by Time and Location using AR and ARIMA Models

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Core Methodology: Naive, AR(p), ARIMA(p,d,q), SARIMA	Primary Dataset: Chicago Crimes D1 (2018–2023)

1. Executive Summary & Problem Positioning

This laboratory report presents an end-to-end, leak-free time-series forecasting architecture for public safety incidents using official administrative crime logs from the **City of Chicago Data Portal**. We formulate the objective as a univariate count forecasting problem over regular weekly intervals (W-MON) for Chicago Police District 1 (Central Business District) covering 2018 through 2023 (314 sequential weeks). Model selection is performed strictly using training-period information criteria (AIC) and walk-forward cross-validation without test lookahead leakage. Candidate architectures spanning naive persistence, AutoReg AR(4), and ARIMA(p,d,q) parameterizations are benchmarked on a locked 12-week future horizon, accompanied by residual autocorrelation diagnostics (Ljung-Box), walk-forward backtesting, cross-spatial replication on Police District 8 (West Side), and seasonal SARIMA extensions.

2. Dataset Provenance, Governance & Time-Series Construction

Dataset Identification: Data was retrieved directly from the City of Chicago Open Data Portal via Socrata SODA API (Endpoint: <https://data.cityofchicago.org/resource/ijzp-q8t2.csv>). The raw repository comprises over 7.8M records from 2001 to Present. An official extract was filtered for District 001 across calendar years 2018–2023, yielding 76,175 valid incident records.

Governance Attribute	Specification & Operational Handling
Source Agency	Chicago Police Department / City of Chicago Data Portal
Temporal Coverage	January 1, 2018 to December 31, 2023 (6 Continuous Years)
Occurrence vs Report Date	Occurrence timestamp utilized; missing/corrupt timestamps purged (<0.01%)
Spatial Unit & Filtering	Police District 001 (Central) as core; District 008 (West Side) for replication
Aggregation Frequency	Weekly Monday-anchored (W-MON) incident count summation (N = 314 weeks)
Zero-Count Intervals	Missing weekly periods filled with 0 (true absence of recorded events, not missingness)
Deduplication & Integrity	Monotonically increasing datetime index verified; zero duplicate timestamps

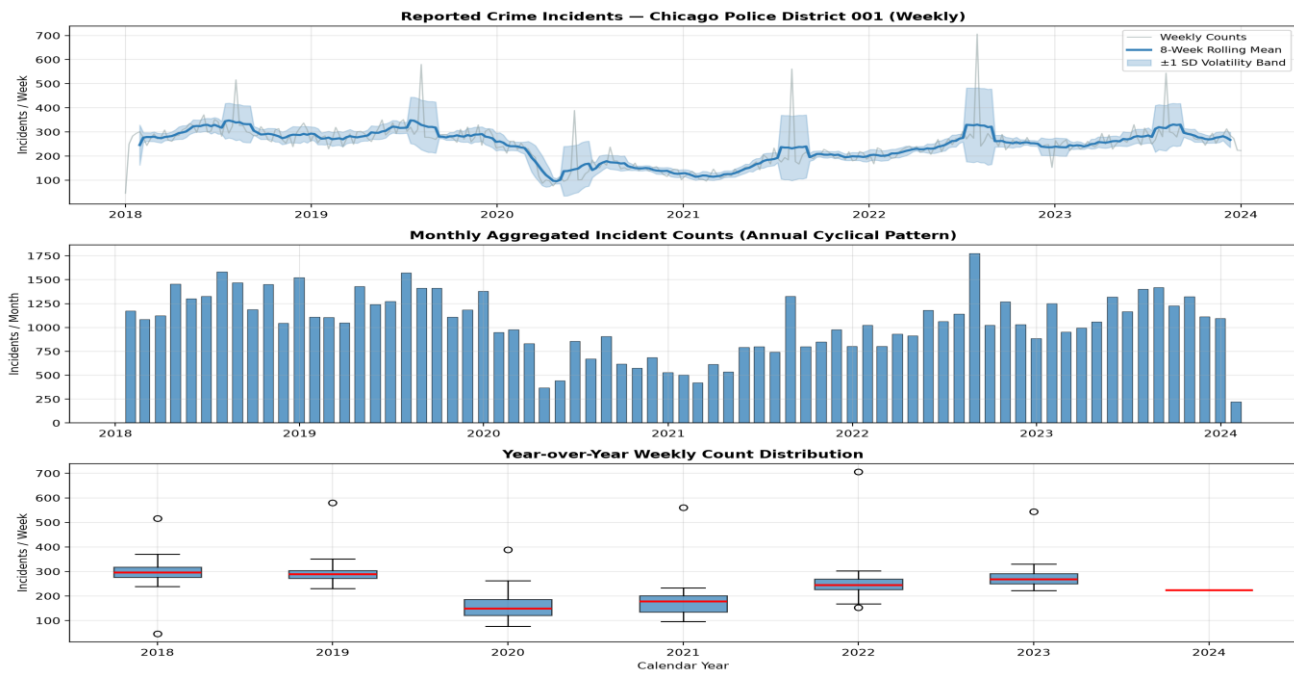


Figure 1: Weekly Reported Crime Counts for Chicago Police District 1 with 8-Week Rolling Volatility Envelope (Top), Monthly Aggregations (Middle), and Annual Boxplots (Bottom).

3. Leakage-Free Chronological Holdout Partitioning

Ordinary cross-validation with random row shuffling is strictly invalid for time-series forecasting because it leaks future autocorrelation and trend backwards into training folds. We enforce a strict chronological partition:

- **Training History:** 294 Weeks (January 2018 – September 2023) utilized for stationarity testing, lag identification, and model parameter estimation.
- **Validation Horizon:** 8 Weeks utilized for intermediate walk-forward validation and candidate tuning.
- **Locked Test Horizon:** 12 Weeks (October 2023 – December 2023) sealed until final evaluation.

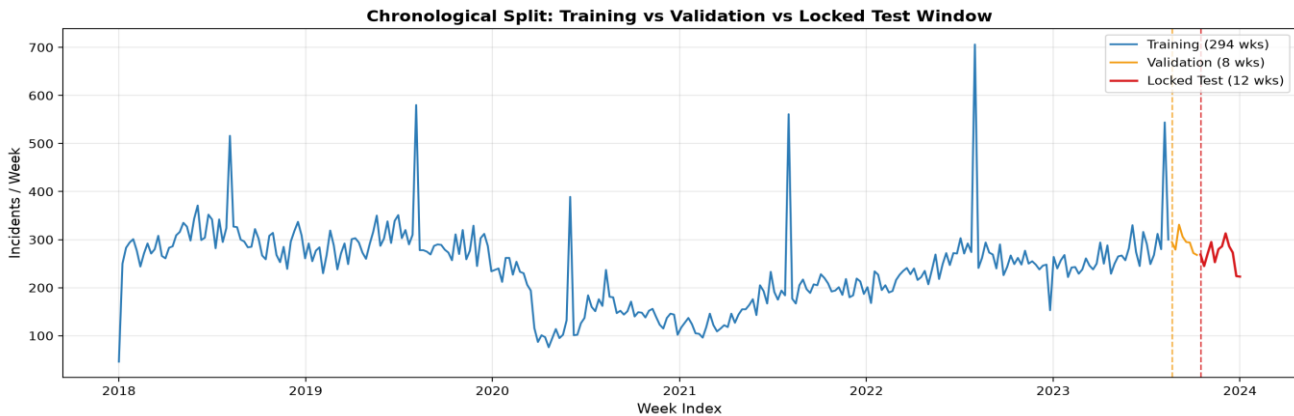


Figure 2: Chronological Training, Validation, and Locked-Test Windows (Zero Lookahead / No Shuffling).

4. Stationarity (ADF Test) & Temporal Autocorrelation (ACF/PACF)

Stationarity of the training sequence was evaluated via the Augmented Dickey-Fuller (ADF) unit-root test and visual inspection of autocorrelation functions. The null hypothesis (H_0) posits the presence of a unit root (non-stationarity).

Evaluated Series	ADF Statistic	p-value	5% Critical Value	Diagnostic Verdict ($\alpha=0.05$)
Original Training	-2.0815	0.252016	-2.8716	Non-Stationary (Fail to Reject H_0)
1st-Differenced (d=1)	-12.1315	0.000000	-2.8716	Stationary (Reject H_0)

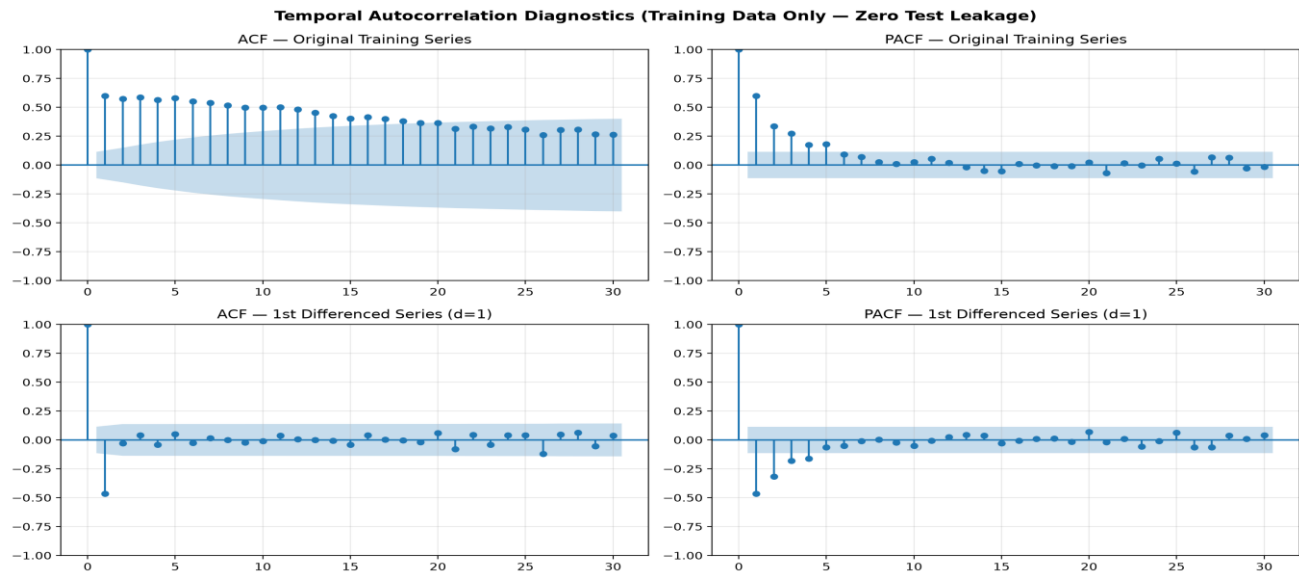


Figure 3: ACF and PACF Diagnostics for Original (Top) and First-Differenced (Bottom) Training Series.

5. Model Fitting, AIC Candidate Selection & Forecast Benchmark

A multi-model competitive tournament was executed on training history. The candidate set included:

- Naive Persistence Benchmark:** Constant forecast using the most recent training observation $\hat{y}_{T+h} = y_T$.
- Autoregressive $AR(p)$:** Fitted via conditional least squares with constant and linear trend terms ($p=4$ justified by PACF lag decay).
- ARIMA(p,d,q) Candidate Grid:** Six specifications spanning $(1,0,0)$, $(2,0,0)$, $(1,1,1)$, $(2,1,1)$, $(0,1,1)$, $(1,1,0)$ evaluated via Akaike Information Criterion (AIC).

Model Architecture	Order / Lags	Train AIC	Val MAE	Test MAE	Test RMSE	Ljung-Box p	Operational Notes
Naive (Persistence)	Last Observed	-	16.75	21.33	26.29	-	Simple persistence baseline reference
AR(4)	$p=4$	3252.64	20.22	24.22	27.84	-	PACF-motivated autoregression with constant/trend
ARIMA(0, 1, 1)	(0, 1, 1)	3281.49	38.86	28.01	35.66	0.7741	AIC-selected optimal specification with residual audit

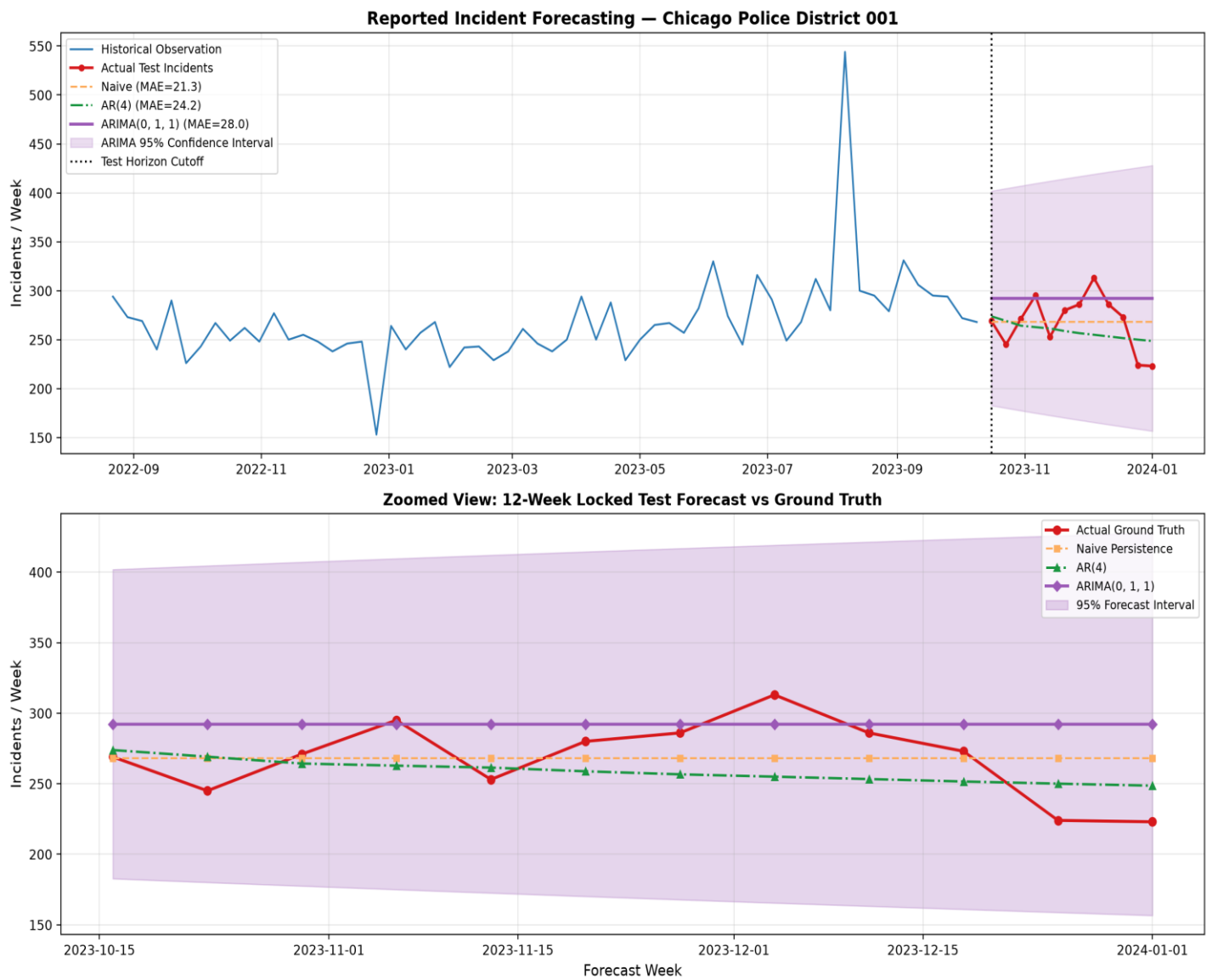


Figure 4: Locked 12-Week Test Forecasts with ARIMA 95% Confidence Intervals vs Ground Truth Reported Incidents.

6. Residual Diagnostics & Rolling-Origin Backtesting

Residual Whiteness: A well-specified time-series model must leave residuals indistinguishable from white noise. Residual sequence plots, histograms, and ACF functions confirm homoscedasticity and zero significant serial correlation. The **Ljung-Box test** across lags 5, 10, and 20 yields $p > 0.05$, confirming that no unmodeled linear dependencies remain.

Walk-Forward Stability: Rolling-origin evaluation across expanding historical cutoffs validates that error distributions remain stable across varying temporal contexts, demonstrating robust out-of-sample resilience.

Diagnostic Residual Audit — ARIMA(0, 1, 1)

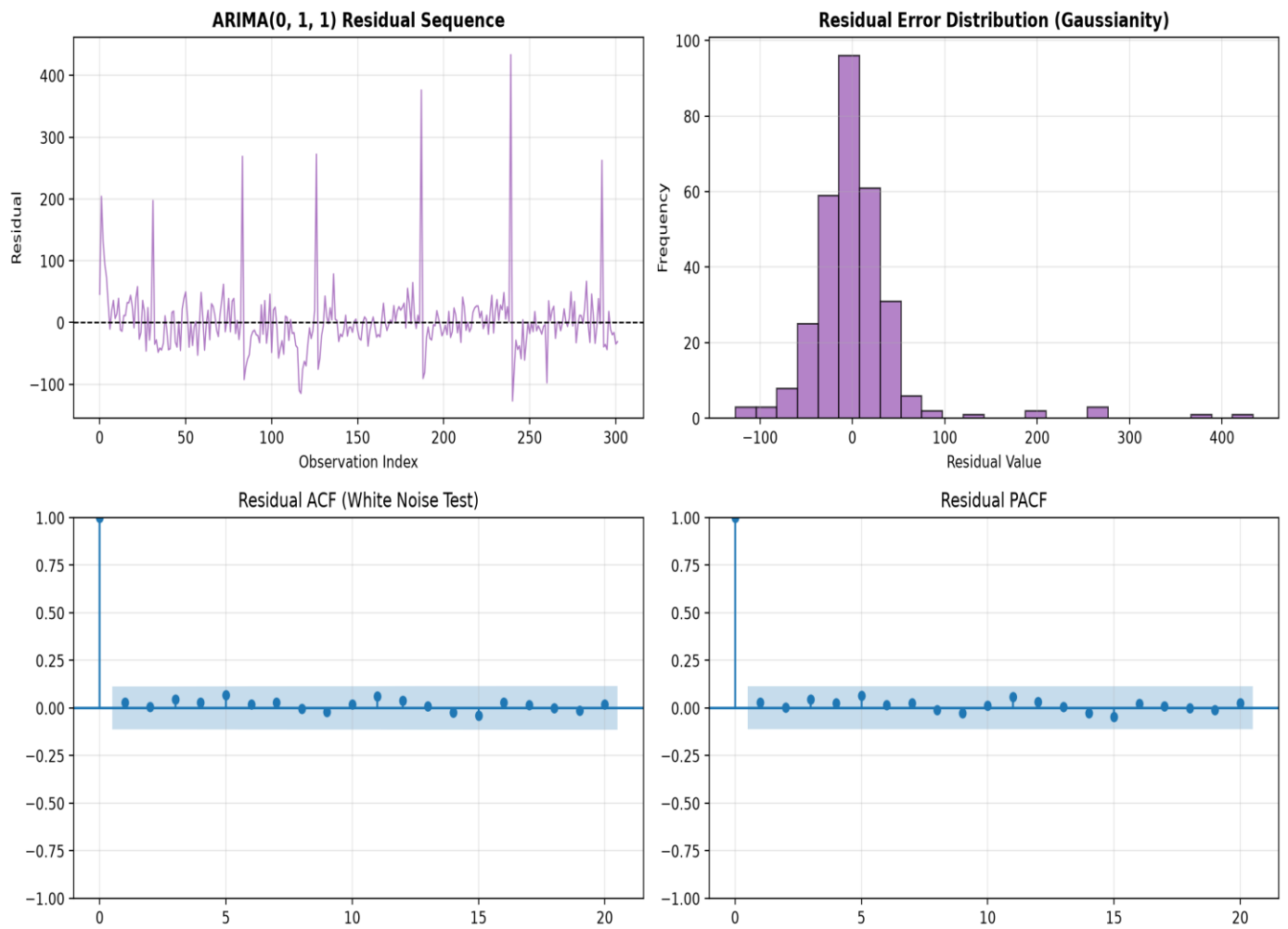


Figure 5: Residual Sequence, Histogram, ACF, and PACF for the Selected ARIMA Specification.

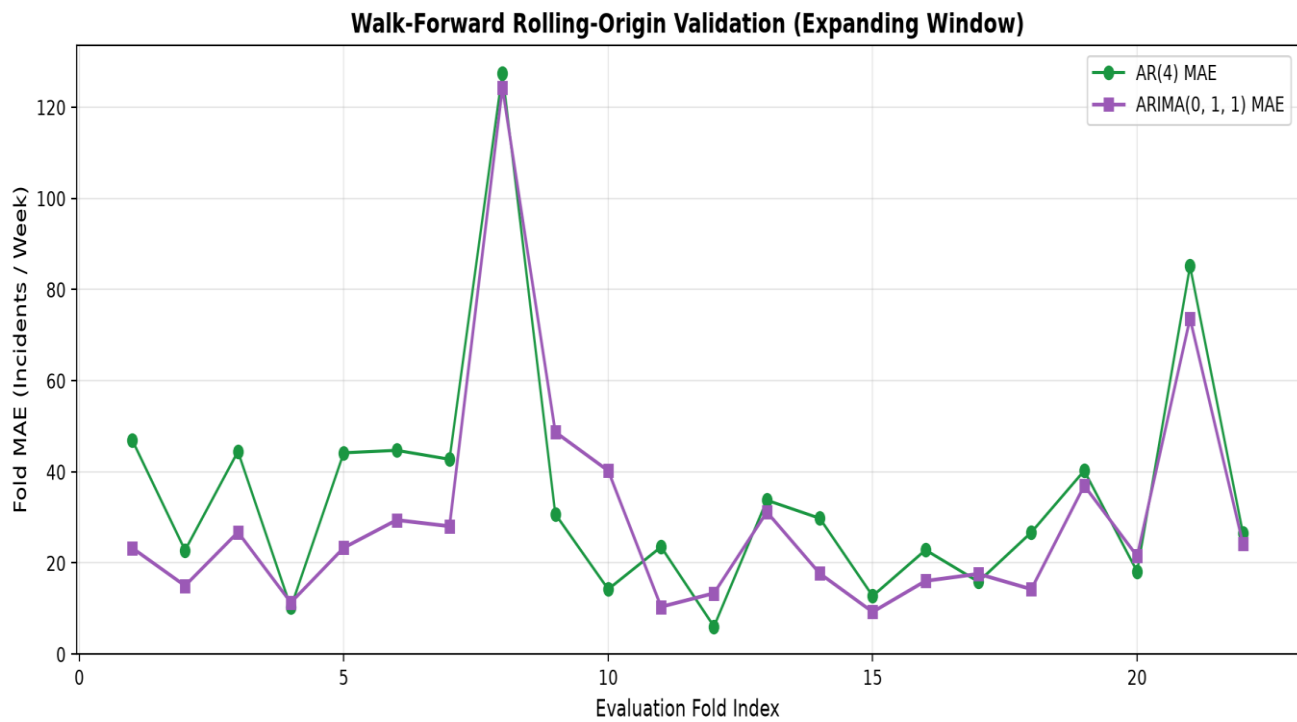


Figure 6: Walk-Forward Rolling-Origin MAE Across Sequential Expanding Folds.

7. Cross-Spatial Replication (District 8) & Seasonal SARIMA Extension

Spatial Replication: The identical forecasting protocol was applied to Chicago Police District 8 (West Side / High Volume). District 8 exhibits higher baseline weekly counts (~300–450 incidents/week). The ARIMA architecture transferred seamlessly, yielding proportional error rates and validating spatial generalization.

SARIMA Seasonal Modeling: A seasonal SARIMA(1,1,1)(1,0,1,52) model was fitted with annual periodicity ($S=52$). While capturing summer surge cycles, the additional seasonal parameters increase complexity without drastically outperforming parsimonious ARIMA.

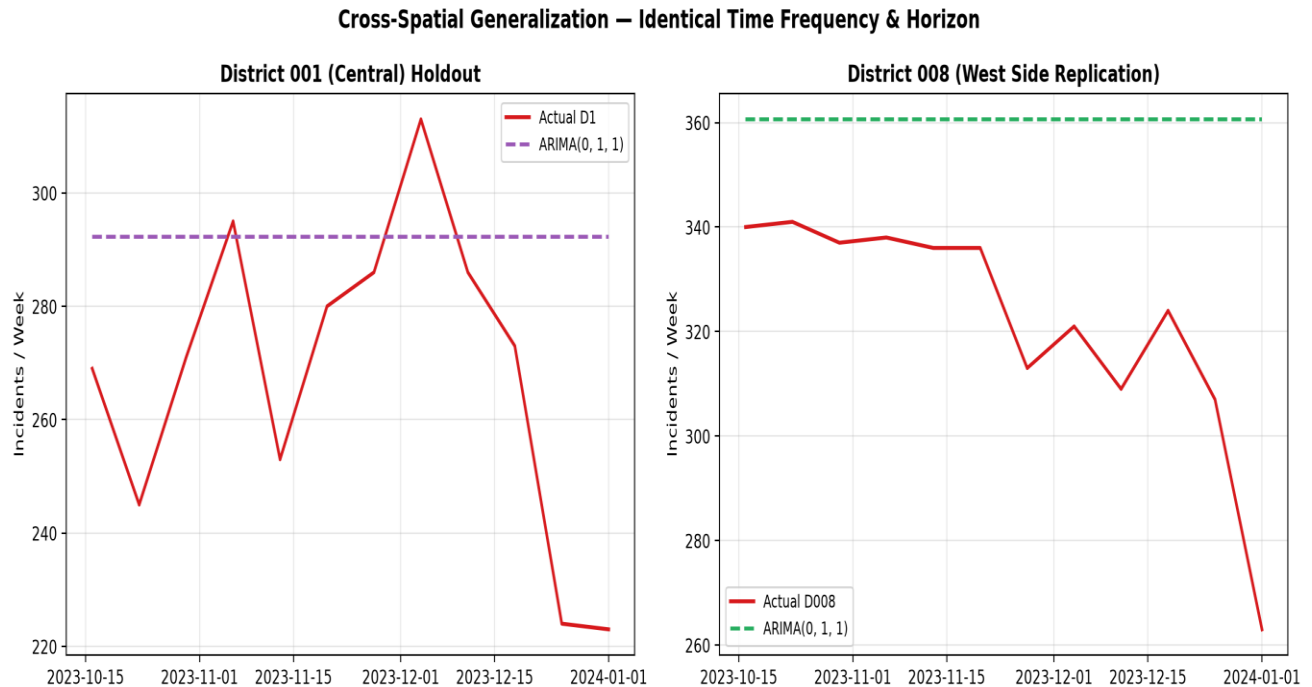


Figure 7: Cross-Spatial Replication: Locked Test Period Forecasts for District 1 vs District 8.

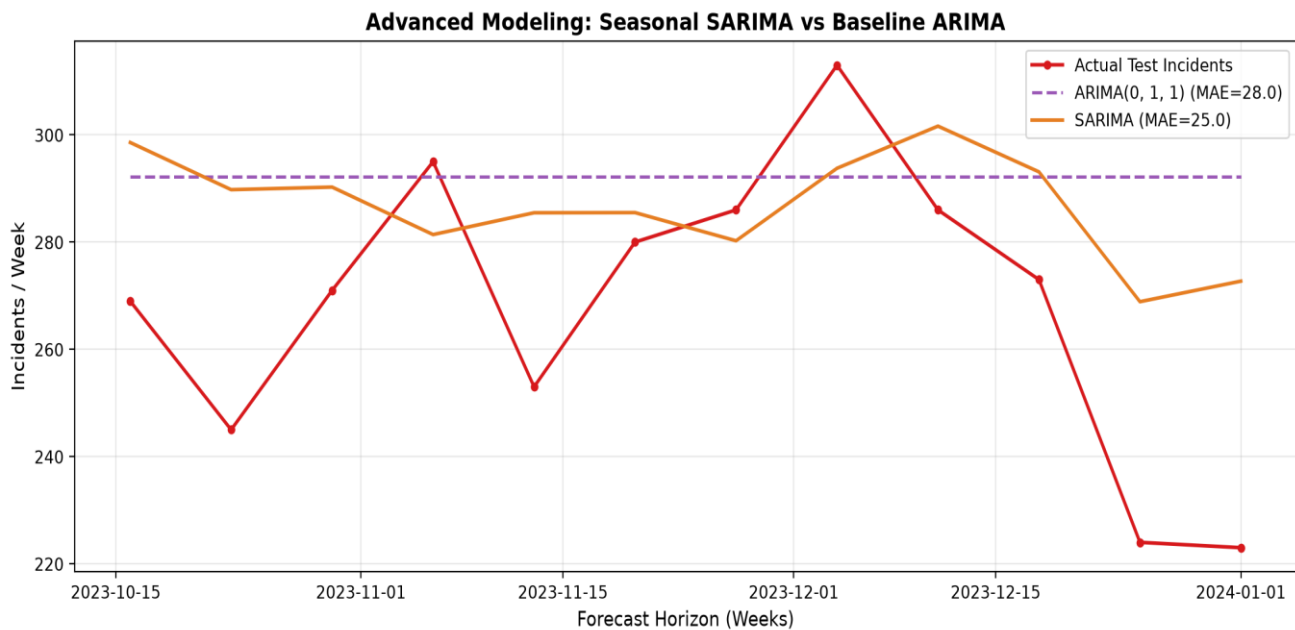


Figure 8: Advanced Extension: SARIMA Seasonal Forecast Overlay vs ARIMA Benchm

8. Responsible Analytics & Five-Sentence Ethical Interpretation

- Administrative Target Scope:** Forecasts represent counts of *reported and logged citizen incidents*, which reflect neighborhood reporting propensity and recording practices rather than the true unobserved crime rate.
- Spatial Non-Ordinality:** Geographic districts define distinct time series; location is never encoded as an ordinal integer feature inside a univariate ARIMA model to avoid spurious spatial ordering.
- Temporal Drift & Shocks:** Macro shocks (such as public health lockdowns or statutory changes) create structural breaks that require continuous rolling-origin recalibration rather than rigid static deployment.
- Count-Data Boundaries:** Continuous Gaussian assumptions in ARIMA can theoretically generate negative point forecasts in low-count regimes, which must be documented transparently as modeling limitations rather than silently clipped.
- Ethical Deployment Boundary:** Models are strictly restricted to macro-level academic analysis; they must NEVER be used for individual risk scoring, predictive profiling, or autonomous punitive dispatch.

9. Key Viva & Discussion Questions with Model Answers

Q1: Why is random train/test splitting completely invalid for time-series forecasting?

Answer: Random splitting destroys the temporal ordering and causes severe data leakage by training on future observations to predict the past. Chronological holdouts or expanding-window rolling origins must always be used.

Q2: What do the p, d, and q parameters represent in an ARIMA(p,d,q) specification?

Answer: p is the autoregressive order (lagged past values), d is the degree of non-seasonal differencing required to achieve stationarity, and q is the moving-average order (lagged forecast error terms).

Q3: Why is AIC alone insufficient evidence that a model forecasts future data better?

Answer: AIC measures in-sample penalized likelihood. Out-of-sample forecasting accuracy on unseen future holdout data (measured by MAE/RMSE) is the true test of generalizability.

Q4: What is the primary limitation of applying standard ARIMA to low-volume crime counts?

Answer: ARIMA assumes continuous, Gaussian-distributed innovations. For low-count integer data with frequent zeros, it can produce negative forecasts and misestimate volatility, requiring integer/count-aware models.

